

The importance of open source AI in IoT

Open source AI is crucial in IoT because it makes powerful AI tools accessible, cost-effective, and customisable for developers. It accelerates the development of AI-driven IoT applications by providing pre-built models and frameworks that can be easily adapted. With strong community support, it fosters innovation and helps solve problems faster. It also enhances interoperability, enabling seamless integration of diverse IoT devices. It optimises edge computing for real-time decision-making and scalability, while ensuring transparency and security. Overall, open source AI empowers businesses to build smarter, more efficient IoT solutions at lower costs.

Open source AI frameworks for IoT

TensorFlow Lite

TensorFlow Lite is designed to enable machine learning (ML) models to run efficiently on mobile, embedded, and edge devices with limited computational resources. Its focus is optimising models for low-latency inference and low power consumption on resource-constrained hardware such as smartphones, microcontrollers, and IoT devices.

Strengths	Weaknesses
Enables AI processing directly on edge devices without cloud dependency.	Converting large TensorFlow models to TensorFlow Lite format can be complex and may result in loss of accuracy.
Uses techniques like quantization, pruning, and reduced precision to reduce model size and improve speed.	May struggle with larger, more complex models that require significant computational resources.
Compatible with Android, iOS, and microcontroller-based devices.	Some IoT devices may require custom optimisations, increasing development time and complexity.
Supports GPUs, DSPs, and specialised AI chips to enhance performance.	Achieving optimal performance on different devices requires specific knowledge of hardware and model tuning.

TinyML

TinyML focuses on running machine learning models directly on small, low-power devices like microcontrollers and IoT sensors, enabling real-time, on-device AI processing. Its main goal is to make machine learning possible in environments where computing resources, memory, and power are extremely limited.

Strengths	Weaknesses
Optimised for ultra-low-power devices, enabling AI in battery-operated IoT devices.	Limited model complexity, simplified models, limiting use in complex AI tasks.
Enables real-time data processing and decision-making at the edge, without cloud dependency.	Memory and storage limitations restrict the size and complexity of models that can be deployed.
Allows data processing at the edge, improving privacy, security, and reducing latency.	Requires specialised knowledge to optimise models for highly constrained devices.
Makes smart features available on low-cost microcontrollers and IoT devices.	Small devices may lack the processing power required for more sophisticated AI models.

Keras

Keras is a high-level deep learning framework that allows easy and fast prototyping for AI models. Integrated with TensorFlow, it simplifies the process of building and deploying deep learning models. Keras is suitable for IoT devices that require lightweight AI models for applications like prediction, anomaly detection, and decision-making.

Strengths	Weaknesses
Makes it easy to design, train, and test neural networks with minimal code.	Higher-level abstractions may not provide the control needed for very custom solutions.
Seamless integration with TensorFlow for enhanced capabilities and tools.	Requires significant computational power, limiting its use in highly constrained devices.
Offers access to a wide variety of pre-trained models for fast deployment.	Running Keras models on constrained devices can lead to performance issues.
Backed by a large open source community with many resources and tutorials.	Not optimised for the smallest IoT devices and edge computing.

Edge Impulse

Edge Impulse is a platform designed for building and deploying machine learning models on embedded systems.